RavenStack SaaS Customer Churn Intelligence

Business Analysis Case Study

Identifying Customer Retention Risks,

Revenue Leakage & Business Growth Opportunities

Prepared By

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Tools Used: Excel | MySQL | Power BI | GitHub

Dataset: RavenStack SaaS Customer Churn Dataset

Source: Kaggle

Dataset Summary:

Domain : SaaS / B2B Customer Analytics

Records : ~500 Customer Accounts

Tables : 5 Relational Tables

Time Period : Jan 2023 – Dec 2024

Executive Summary

RavenStack is a B2B SaaS company preparing for public launch. Before scaling its business, the leadership team wanted to understand why customers were leaving, which customer segments were most at risk, and how churn was affecting revenue.

This project analyzes customer subscriptions, support interactions, product usage, refunds, and account information to identify the business drivers behind customer churn. The analysis was carried out using Excel for data exploration, SQL for business analysis, and Power BI for executive reporting.

The findings show that churn is not random. DevTools customers account for the highest churn volume, paid acquisition channels perform worse than organic channels, and most customers leave after using the product for more than 90 days. The Pro plan contributes the highest revenue loss, while feature gaps emerge as the leading reason behind customer refunds.

Based on these findings, the report recommends strengthening long-term customer engagement, prioritizing retention efforts for high-value Pro customers, improving acquisition quality, and addressing the product gaps driving customer dissatisfaction. Together, these actions can reduce churn, protect revenue, and support RavenStack's long-term growth.

Project Overview

Customer retention is one of the most important growth indicators for any SaaS business. While acquiring new customers requires significant investment, retaining existing customers directly improves revenue, customer lifetime value, and long-term business sustainability.

This project investigates customer churn at RavenStack, a B2B SaaS company preparing for public launch. Using historical customer data collected during its pilot phase, the analysis identifies the customer segments, business factors, and operational challenges contributing to customer churn.

The project follows a structured Business Analytics approach. Customer data was first explored in Excel to understand quality and patterns. SQL was then used to answer key business questions and uncover churn drivers. Finally, Power BI transformed the findings into an executive dashboard that supports faster and more informed business decisions.

The final outcome is not only an interactive dashboard but also a business case study that explains the causes of churn, quantifies its impact on revenue, and provides practical recommendations to improve customer retention before product launch.

Project Scope

- Analyze customer churn across industries, acquisition channels, subscription plans, and product usage.

- Measure the financial impact of churn through revenue loss analysis.
- Identify customer behavior patterns that influence retention.
- Present findings through an executive-level Power BI dashboard.
- Recommend business actions to reduce churn and improve long-term customer retention.

Tools Used

- **Microsoft Excel** – Data exploration and exploratory analysis
- **MySQL** – Business analysis and query-based insights
- **Power BI** – Dashboard development and executive reporting

Business Scenario & Problem Statement

RavenStack is a B2B SaaS startup preparing to launch its AI-powered collaboration platform to a wider market. During its pilot phase, the company acquired customers across multiple industries, subscription plans, and acquisition channels while tracking subscriptions, feature usage, support interactions, refunds, and customer churn.

Although the pilot generated valuable customer data, leadership lacked visibility into why customers were leaving and which business areas required immediate attention. Churn was increasing, refunds were affecting revenue, and support teams were handling a growing number of customer issues. Without understanding the root causes behind these trends, expanding the customer base would increase acquisition costs without improving long-term growth.

RavenStack needed a structured business analysis to answer four critical questions:

- Which customer segments are most likely to churn?
- What factors are driving customer churn and revenue loss?
- At what stage of the customer journey does churn occur?
- Which business actions should be prioritized before scaling the product?

This project was undertaken to transform raw customer data into business insights that help leadership reduce churn, protect revenue, and improve customer retention before the company's public launch.

Business Objectives

The objective of this project is to identify the key factors influencing customer churn and measure their impact on revenue, customer retention, and business performance. The analysis aims to uncover high-risk customer segments, evaluate acquisition channel effectiveness, understand when customers are most likely to leave, identify the primary reasons behind refunds, and provide practical recommendations that help RavenStack improve customer retention before expanding its customer base.

Stakeholder Analysis

Stakeholder	Business Need
CEO / Leadership Team	Monitor customer churn, revenue leakage, and overall business health to support strategic growth decisions.
Product Manager	Identify product gaps and customer pain points to improve feature adoption and long-term retention.
Customer Success Team	Detect high-risk customer segments and prioritize proactive retention initiatives.
Marketing Team	Evaluate acquisition channel performance and improve customer quality across marketing campaigns.
Sales Team	Understand which industries and subscription plans generate long-term customer value.
Support Team	Analyze support trends and customer issues that contribute to churn and refund requests.

Dataset Overview

The analysis is based on five interconnected datasets that represent RavenStack's customer lifecycle—from account creation to subscription activity, product usage, customer support, and eventual churn. Together, these datasets provide a complete view of customer behavior and enable end-to-end churn analysis.

Dataset	Business Purpose
Accounts	Stores customer profile information, including industry, country, acquisition source, and company details. Used for customer segmentation.
Subscriptions	Contains subscription plans, billing details, renewal status, and churn information. Used to measure customer retention, plan performance, and revenue impact.
Feature Usage	Records customer interaction with product features, usage frequency, session duration, and product errors. Used to understand product adoption and customer

Dataset	Business Purpose
	engagement.
Support Tickets	Captures customer support requests, response times, escalation status, and satisfaction scores. Used to evaluate customer experience and operational performance.
Churn Events	Stores churn dates, refund amounts, refund reasons, and customer lifetime information. Used to identify churn timing, financial impact, and the primary reasons behind customer attrition.

Dataset Summary

- **Total Customers:** 500
- **Datasets Used:** 5
- **Analysis Period:** Pilot phase before public launch
- **Primary Business Domain:** B2B SaaS Customer Retention & Churn Analytics

Business Questions

The analysis was designed to answer the following business questions before RavenStack's public launch:

1. Which customer segments contribute the highest churn, and where should retention efforts be prioritized?
2. How do acquisition channels, subscription plans, and customer behavior influence churn and revenue loss?
3. At what stage of the customer lifecycle are customers most likely to leave the platform?
4. What operational and product-related factors are driving customer dissatisfaction and refund requests?
5. What business actions can reduce churn, protect recurring revenue, and improve long-term customer retention?

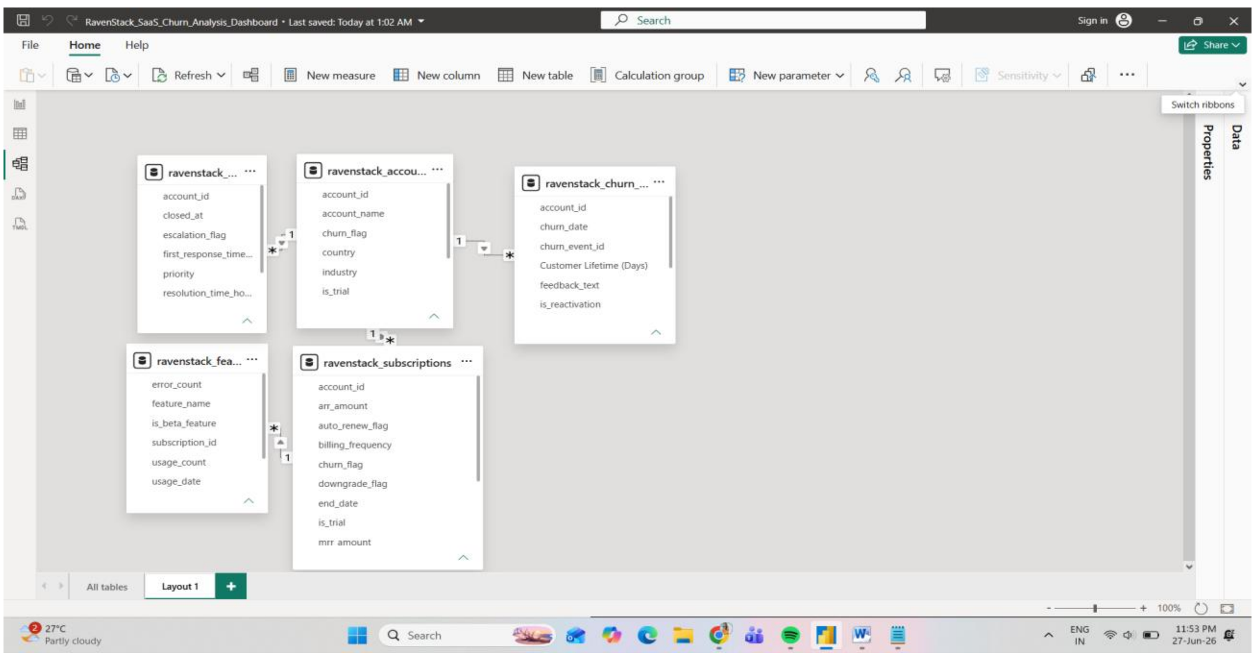
Key Takeaway:
These questions guided the entire analysis and formed the foundation for the Excel exploration, SQL business analysis, Power BI dashboard, and final business recommendations.

Data Preparation & Modeling

1. Data Preparation

Activity	Purpose
Data Validation	Verified row counts, data types, and missing values across all datasets.
Relationship Mapping	Connected five business datasets using `account_id` as the primary key.
Data Cleaning	Removed inconsistencies and validated categorical fields before analysis.
Feature Engineering	Created business metrics such as Customer Lifetime, Revenue Loss, Churn Rate, and Time-to-Churn.
Data Modeling	Built a relational model to support cross-functional business analysis.

2. Data Model



The data model connects five business entities through **account_id**, creating a unified customer view across subscriptions, feature usage, support interactions, and churn events. This relationship enables end-to-end analysis of customer behavior throughout the product lifecycle.

3. Business Metrics Created

Business Metric	Business Purpose
Churn Rate	Measures percentage of customers who discontinued their subscription.
Revenue Loss	Quantifies recurring revenue lost due to churn.
Customer Lifetime	Calculates the number of days a customer remained active before churn.
Time-to-Churn Group	Segments customers into lifecycle stages to identify churn timing.

4. Key Takeaway

A clean relational data model and business-focused metrics created a reliable foundation for analyzing customer behavior, identifying churn drivers, and measuring revenue impact.

Exploratory Data Analysis (Excel)

The first stage of analysis focused on understanding the overall customer landscape and identifying early business patterns before performing deeper SQL analysis. Excel was used to validate the datasets, summarize customer behavior, and highlight potential churn indicators through pivot tables and descriptive analysis.

Key Analysis Performed

- Customer distribution across industries, countries, and subscription plans.
- Churn rate comparison across customer segments.
- Revenue loss by subscription plan.
- Customer acquisition analysis by referral source.
- Refund trends by refund reason.
- Initial validation of support and subscription metrics.

Major Findings

- DevTools represented the largest share of churned customers.
- Paid acquisition channels showed weaker customer retention than organic channels.
- The Pro plan generated the highest revenue loss despite similar churn rates.
- Product feature limitations emerged as the leading refund reason.
- Initial analysis suggested that churn was driven by multiple business factors rather than a single operational issue.

Key Takeaway:
Excel provided the first business view of customer behavior, helping identify the major churn patterns that were later validated through SQL and visualized in the executive Power BI dashboard.

Revenue by Plan Tier (MRR & ARR)

Row Labels	Sum of mrr_amount	Sum of arr_amount
Basic	760437	9125244
Enterprise	8473221	101678652
Pro	2105089	25261068
Grand Total	**11338747**	**136064964**

Upgrade_Analysis

Row Labels	Count of subscription_id
FALSE	4471
TRUE	529
Grand Total	**5000**

Downgrade_Analysis

Row Labels	Count of subscription_id
FALSE	4782
TRUE	218
Grand Total	**5000**

Billing Frequency Analysis

Row Labels	Count of subscription_id	Sum of arr_amount
annual	2461	67168776
monthly	2539	68896188
Grand Total	**5000**	**136064964**

Churn_events | EDA_Accounts | EDA_Subscriptions | EDA

Churn Reason Analysis

Row Labels	Count of churn_event_id
features	114
support	104
budget	104
unknown	95
competitor	92
pricing	91
Grand Total	**600**

Refund Analysis by Churn Reason

Row Labels	Sum of refund_amount_usd
features	1905.9
unknown	1742.39
pricing	1333.25
budget	1247.54
support	1219.64
competitor	1203.53
Grand Total	**8652.25**

Reactivation Analysis

Row Labels	Count of churn_event_id
FALSE	539
TRUE	61
Grand Total	**600**

Ticket Priority Distribution

Row Labels	Count of ticket_id
high	510
low	485
medium	491
urgent	514
Grand Total	**2000**

Satisfaction Score Analysis

Row Labels	Count of ticket_id
3	396
4	405
5	374
(blank)	825
Grand Total	**2000**

Escalation Analysis

Row Labels	Count of ticket_id
FALSE	1905
TRUE	95
Grand Total	**2000**

Resolution Time Analysis

Row Labels	Average of resolution_time_hours
high	36.95882353
low	36.34639175
medium	35.59470468
urgent	34.56809339
Grand Total	**35.861**

SQL Business Analysis

SQL was used to investigate the business questions identified during the exploratory analysis. While Excel highlighted initial customer patterns, SQL enabled deeper analysis by combining multiple business datasets and measuring the relationship between customer behavior, churn, revenue, product usage, and operational performance.

Business Questions Addressed

- Which industries contribute the highest customer churn?
- Which acquisition channels generate long-term customers?
- Which subscription plans result in the highest revenue loss?
- When are customers most likely to churn?
- What are the primary reasons behind customer refunds?
- How do support interactions influence customer retention?

Key Findings

- DevTools customers accounted for the highest volume of churned accounts.
- Paid acquisition channels showed higher churn than organically acquired customers.
- The Pro subscription generated the greatest revenue loss despite maintaining churn rates comparable to other plans.
- Most customers exited after remaining active for more than 90 days.
- Missing product features were the leading cause of customer refunds.
- Support-related metrics showed limited influence on churn when compared to product-related issues.

Business Impact

The SQL analysis confirmed that customer churn was influenced by a combination of customer profile, acquisition quality, product experience, and subscription value rather than a single operational factor. These findings established the analytical foundation for the executive Power BI dashboard and the final business recommendations.

Key Takeaway:
SQL transformed raw customer data into measurable business evidence, allowing churn drivers, revenue leakage, and customer risk patterns to be validated before presenting them through executive reporting.

SQL Investigation Summary

Business Question	SQL Outcome
Highest churn segment	DevTools Industry
Weakest acquisition source	Paid Ads
Highest revenue loss	Pro Plan
Customer lifecycle risk	90+ Days
Top refund driver	Missing Features
Operational observation	Support had lower impact than product quality

PowerBI Report

The final deliverable of this project is an interactive executive dashboard built in Power BI. It consolidates customer, subscription, support, and churn data into a single business view, enabling leadership to monitor customer retention, revenue impact, and high-risk business segments without reviewing multiple reports.

The dashboard was designed using an executive-first approach. Instead of presenting large volumes of operational data, it focuses on the business metrics and insights that directly influence strategic decisions. Interactive slicers allow stakeholders to analyze churn across different countries, industries, subscription plans, and customer cohorts.

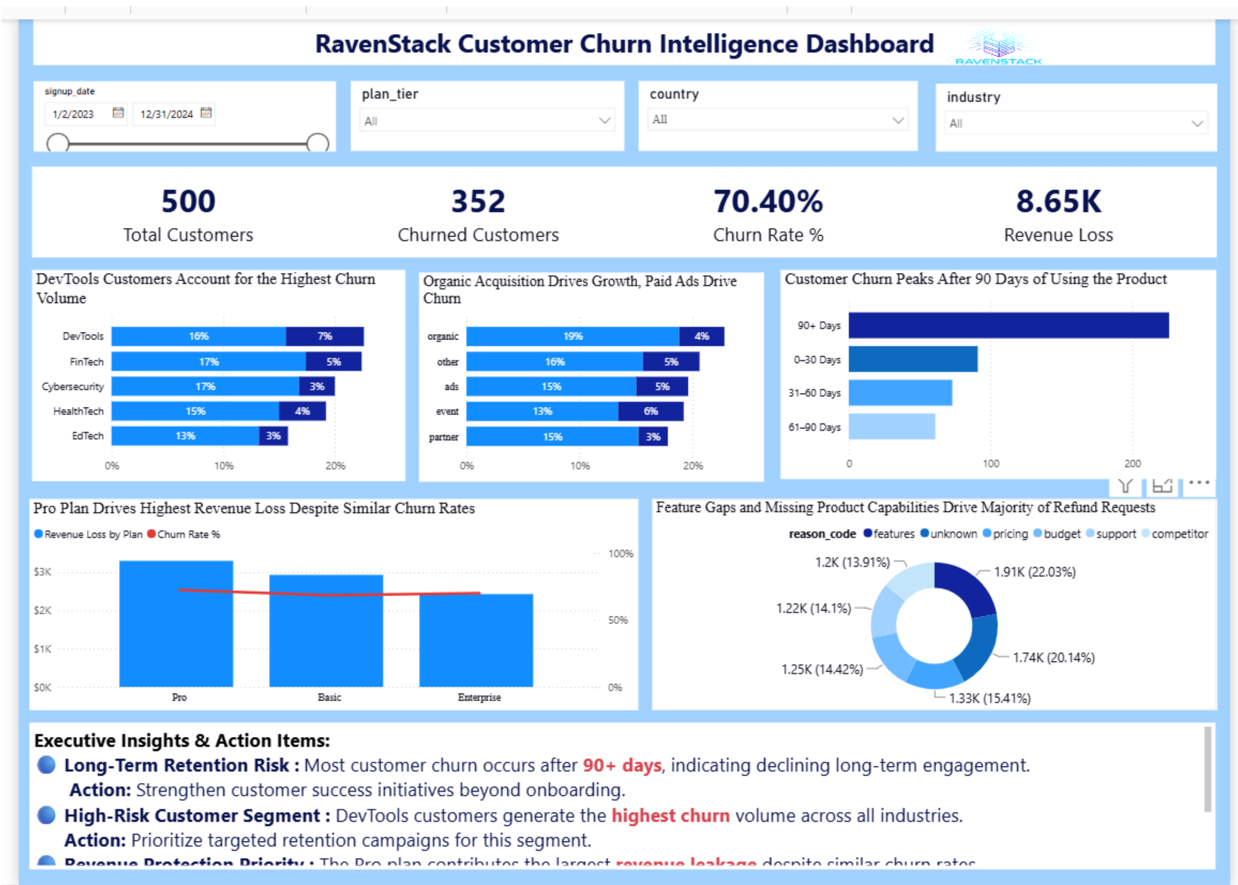
Dashboard Components

Dashboard Section	Business Purpose
KPI Summary	Provides an instant view of total customers, churn rate, revenue loss, and refund value.
Customer Segmentation	Identifies the industries and acquisition channels contributing the highest churn.
Customer Lifecycle	Shows when customers are most likely to leave the platform.
Revenue Impact	Measures revenue loss across subscription plans and compares it with churn rates.
Refund Analysis	Highlights the leading business reasons behind customer refunds.
Executive Insights	Summarizes the key findings and recommended business actions for leadership.

Dashboard Features

- Interactive filtering by **Country**, **Industry**, **Plan Tier**, and **Signup Date**.
- Business-focused visualizations designed for executive decision-making.
- Consistent KPI definitions across all visuals.
- Insight-driven chart titles that communicate business findings rather than chart descriptions.

Key Takeaway:
The dashboard transforms analytical findings into a single executive decision-making platform, allowing RavenStack's leadership to quickly identify customer risks, understand revenue impact, and prioritize retention initiatives.



Business Insights

1. Customer Retention Risk Extends Beyond Onboarding

Although early-stage churn exists, the largest volume of customer exits occurs after **90+ days** of product usage. This indicates that RavenStack's primary challenge is sustaining long-term customer engagement rather than acquiring new users.

Business Impact: Customer lifetime value declines despite successful customer acquisition.

Recommendation: Strengthen customer success programs, introduce proactive engagement campaigns, and monitor product adoption beyond the onboarding phase.

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2. DevTools Represents the Highest Business Risk

The DevTools segment contributes the largest share of total churn volume. Since it also represents RavenStack's largest customer base, even small improvements in retention could produce significant business impact.

Business Impact: Revenue leakage is concentrated within the company's most valuable customer segment.

Recommendation: Prioritize DevTools customers with targeted retention initiatives, dedicated account management, and product-specific engagement strategies.

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3. Organic Acquisition Delivers Better Customer Quality

Organic channels contribute the largest customer base, while paid acquisition channels show comparatively higher churn volume.

Business Impact: Customer acquisition quality differs across marketing channels, affecting long-term retention and acquisition ROI.

Recommendation: Increase investment in high-performing organic acquisition strategies while optimizing paid campaigns for customer quality rather than acquisition volume.

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4. Revenue Risk Is Driven by High-Value Customers

The Pro subscription plan generates the highest revenue loss despite maintaining churn rates similar to other plans.

Business Impact: Losing premium customers creates a disproportionate financial impact on the business.

Recommendation: Implement premium customer success programs, proactive renewal strategies, and early warning alerts for high-value accounts.

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5. Product Experience Drives Refund Requests

Refund requests are primarily associated with missing product capabilities rather than pricing concerns.

Business Impact: Product limitations directly influence customer dissatisfaction and revenue recovery.

Recommendation: Prioritize feature development based on customer feedback before introducing pricing or promotional changes.

Conclusion

This project demonstrates how business analysis can transform raw SaaS operational data into actionable business decisions. By combining Excel for exploratory analysis, SQL for business investigation, and Power BI for executive reporting, the analysis identified the key drivers of customer churn, revenue loss, and customer dissatisfaction.

Rather than focusing solely on reporting metrics, the project delivers practical recommendations that support customer retention, improve product strategy, and protect recurring revenue. The final dashboard enables business leaders to interactively explore customer behavior, evaluate business risks, and prioritize high-impact retention initiatives through a single executive view.

Overall, this case study reflects an end-to-end Business Analyst workflow—from business understanding and data preparation to insight generation and executive decision support.